

Requirements Specification Document (RSD)

1. Introduction

Local governments struggle to track and resolve civic issues (potholes, garbage, broken streetlights). Citizens lack a reliable way to report and follow up. Current solutions only support reporting, not accountability and gamification. Goal: Build a mobile-first PWA and web-based admin dashboard that empowers citizens and government staff.

2. Stakeholders

- Citizens → Report & track civic issues - Department Admins → Receive & resolve complaints - Higher Authorities → Handle escalations, monitor performance - Government → Ensure compliance and analyze trends

3. Functional Requirements

Citizen Features: Register/Login, Submit complaint (photo+location+desc), Track status, Review, Escalate unresolved issues, Earn rewards. Admin Features: Dashboard, Filter/Assign complaints, Update progress, Communicate with citizen. Higher Authority Features: Handle escalations, Monitor performance, Analytics. Gamification: Points redeemable for incentives. Escalation & Confirmation: Time-bound resolution, Auto-close after 72 hrs if unconfirmed.

4. Non-Functional Requirements

Performance, Scalability, Security, Usability, Accessibility (multilingual + offline PWA).

5. Technical Requirements

Frontend: React.js (PWA), Tailwind, Maps API. Backend: Node.js, Express, Clerk (Auth). Database: PostgreSQL/MongoDB. Storage: AWS S3/Firebase. Notifications: Firebase Push, Email, SMS. Admin Dashboard: React.js web app with RBAC. AI Module: Auto-categorization of complaints.

6. Data Requirements

Users Table, Complaints Table, Rewards Table, Escalations Table.

7. Constraints

- Must work on low internet bandwidth (rural areas) - Prototype must be built in hackathon time limit (36–48 hrs) - Must comply with Indian data privacy rules

8. Risks & Challenges

Fake/duplicate complaints, Citizens not confirming resolutions, Resistance from govt staff, Server overload. Mitigation: OTP login, AI duplicate detection, Auto-close, Cloud auto-scaling.

9. Future Scope

AI-based issue severity detection, UPI reward redemption, Blockchain transparency, Open data portal.

10. Business Model

Government funded. Rewards sponsored via govt schemes, corporate CSR, local services. Analytics subscription for planners.

11. Why Better than Existing Systems

Gamification, Escalation, Confirmation, Transparency → ensures accountability beyond reporting.